

X-RAY GÖRÜNTÜLERİNDEN AKCİĞER HASTALIKLARI TAHMİNİ

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Özetçe —Günümüzde yapay zeka ve derin öğrenme tekniklerinin tıp alanındaki kullanımı hızla yaygınlaşmış ve algoritma destekli tanı sistemlerinin geliştirilmesi kritik bir önem kazanmıştır. Bu projede, göğüs X-ray görüntüleri kullanılarak akciğer hastalıklarının tespiti ve sınıflandırılması üzerine bir çalışma gerçekleştirilmiştir. Çalışmanın amacı doğrultusunda, literatürde başarısı kanıtlanmış hem CNN hem de ViT tabanlı mimariler ele alınmıştır. Bu kapsamda InceptionV3, ResNet50, VGG16, ChexNet, SwinT ve ViT-B16 modelleri kullanılmıştır. Eğitim sürecinde modellerin yeteneğini artırmak amacıyla veri kümesi üzerinde çeşitli ön işleme ve augmentation teknikleri uygulanmıştır. Modellerin başarısı sadece doğruluk oranıyla değil aynı zamanda kesinlik, duyarlılık, F1 Skoru ve Cohen's Kappa katsayısı gibi oranlarla da değerlendirilmiştir.

Anahtar Kelimeler—Makine öğrenmesi, yapay sinir ağları, akciğer hastalıkları, derin öğrenme algoritmaları

Abstract—Today, the use of artificial intelligence and deep learning techniques in medicine has rapidly increased, and the development of algorithm-supported diagnostic systems has become critically important.

In line with the study's objective, both CNN and ViT-based architectures, proven successful in the literature, were considered. In this context, InceptionV3, ResNet50, VGG16, ChexNet, SwinT, and ViT-B16 models were used. Various preprocessing and augmentation techniques were applied to the dataset during the training process to improve the models' capabilities. The success of the models was evaluated not only by accuracy but also by statistical reliability metrics such as precision, sensitivity, F1 score, and Cohen's Kappa coefficient.

Keywords—Machine learning, neural network, lung diseases, deep learning algorithm

I. INTRODUCTION

In recent years, Deep Learning (DL) techniques, particularly Convolutional Neural Networks (CNN), have been established as the "gold standard" in medical image analysis. Architectures such as Inception, ResNet, and VGG have demonstrated high success rates across various pathologies due to their hierarchical feature extraction capabilities. However, the adaptation of Transformer architectures which revolutionized natural language processing—to computer vision has introduced models like Vision Transformer (ViT) and Swin Transformer. These models offer a powerful alternative to CNN-based approaches by modeling long-range dependencies in images, presenting new opportunities for detecting complex lung pathologies.

In this study, we comparatively analyzed the performance of traditional CNN architectures (InceptionV3, ResNet50, VGG16, ChexNet) and modern Transformer based models (ViTB16, Swin Transformer) on a comprehensive dataset containing five distinct classes: Covid-19, Lung Opacity, Normal, Viral Pneumonia, and Tuberculosis. The primary objective of this study is to evaluate not only the accuracy but also the stability and sensitivity of these models to determine the most suitable architecture for clinical use. Experimental results are discussed in detail through confusion matrices and statistical metrics (F1-Score, Kappa).

II. LITERATURE REVIEW

Early diagnosis of lung diseases is crucial in many ways today. It impacts everything from providing higher quality treatment to offering less expensive treatment. Most importantly, early diagnosis increases survival rates. X-ray images are the most important aids for specialists due to their speed, painlessness, non-surgical application, the fact that air hardly obstructs the X-rays, and their widespread use. The examination of these images by specialists and diagnosis is the next stage [1].

This stage presents certain challenges. These include a shortage of specialists to examine thousands of radiological images, differing interpretations of images among specialists, the similarity of diseases, or the possibility of a diagnosis being overlooked due to factors such as fatigue. Studies have been conducted to address these shortcomings [2], [3].

Disease prediction studies using deep learning models are quite popular today. Unlike traditional machine learning models, it automates the manual feature extraction process – telling the model which features to focus on. All that needs to be done is to feed the labeled data to the model that will predict the disease [4].

In the literature review section, deep learning methods used in various studies and their success rates will be examined, and the most suitable methods will be identified to continue the work in order to achieve high efficiency [5].

III. MATERIAL - METHODS

A. Material

For model training, non-profit datasets shared as open source by Harvard University were used. A team of

researchers from Qatar University, Doha University, and Dhaka University, in collaboration with physicians in Pakistan and Malaysia, created a 4-class (COVID-19, Lung Opacity, Normal, and Viral Pneumonia) dataset. The primary dataset was collected from Kaggle (COVID-19 Radiography Database). Some additional content related to pneumonia and COVID-19 from various online sources was also added. In our research study, an additional class, tuberculosis, was added to the 4-class dataset.

Lateral view chest X-rays were removed from the dataset, and only the frontal view was added. The final dataset created is a 5-class chest X-ray dataset related to infectious respiratory diseases. The dataset contains a total of 32,687 samples. However, not all of these samples were used; some pre-processing was performed. First, rows containing missing values were deleted from the dataset, and then similar examples were also deleted. After the deletions, dataset manipulations were applied to improve the model's performance. The data was presented as test, validation, and training samples. The data was used as a whole. These processes will be detailed in other sections [6].

Table 1 Number of Data

Disease Name	Data Count	Label
Covid-19	4189	0
Lung-Opacity	6012	1
Normal	10192	2
Viral Pneumonia	7397	3
Tuberculosis	4897	4

B. Methods

In this study, we evaluated six distinct deep learning architectures, ranging from established CNN to state of the art Vision Transformers, to determine their efficacy in diagnosing lung diseases. System process detailed in Figure 1.



Figure 1 Block schema of system

- **VGG16:** A classic CNN architecture known for its simplicity and depth. It utilizes a uniform arrangement of small 3×3 convolution filters stacked on top of each other. While computationally intensive, VGG16 is highly effective at extracting hierarchical feature representations from medical images.
- **ResNet50:** To address the vanishing gradient problem in deep networks, ResNet50 introduces "residual learning" through skip connections. These connections allow gradients to flow through the network more easily, enabling the training of deeper architectures without performance degradation.
- **InceptionV3:** This architecture focuses on computational efficiency using "Inception modules"

that apply factorized convolutions and asymmetric filters. This design allows the model to capture features at multiple scales simultaneously with fewer parameters compared to VGG or ResNet.

- **ChexNet:** Based on the DenseNet121 architecture, ChexNet connects each layer to every other layer in a feed-forward fashion (dense connections). It facilitates strong feature propagation and reuse, making it particularly specialized for chest radiograph analysis.
- **ViT-B/16 (Vision Transformer):** Representing a shift from CNNs, ViT treats an image as a sequence of fixed-size patches (16×16 pixels). It utilizes self-attention mechanisms to model long-range dependencies and global context across the entire image, which is often challenging for standard CNNs.
- **Swin Transformer (Swin-T):** A hierarchical Transformer that addresses the computational complexity of ViT. It computes self-attention within non-overlapping local windows and utilizes a "shifted window" mechanism to allow cross-window connection. This approach makes it highly efficient for high-resolution medical imaging.

IV. RESULT AND DISCUSSION

In this study, the performance of six deep learning architectures (InceptionV3, ResNet50, Chexnet, VGG16, Swin Transformer, and ViT-B-16) was evaluated for the classification of lung diseases. The models were assessed using Accuracy, Precision, Recall, F1-Score, and Cohen's Kappa metrics. Furthermore, the standard deviation ($\pm$) was calculated across the 5-fold cross-validation to measure the stability and robustness of each model.

A. Performance Analysis

InceptionV3 demonstrated the highest overall performance among all tested models. As detailed in the experimental results, it achieved an average accuracy of 96.43% and an F1 Score of 96.42%. The low standard deviation (± 0.33) combined with a Kappa score of 0.9541 indicates that the model provides highly consistent and reliable predictions across different data folds. The detailed performance metrics shown Table 2.

Swin Transformer (SwinT) emerged as the strongest competitor to the CNN-based InceptionV3. It achieved a nearly identical accuracy of 96.42% and an F1 Score of 96.41%. With a Kappa score of 0.9539, the SwinT model demonstrated that Transformer-based architectures can achieve "near-perfect" agreement in medical image classification, performing on par with the best CNN models. The detailed performance metrics shown Table 3.

ResNet50 also exhibited robust results with an accuracy of 96.20% and an F1 Score of 96.19%. Notably, ResNet50 showed superior stability compared to InceptionV3, evidenced by a lower standard deviation (± 0.19). This suggests that the residual connections

effectively mitigate overfitting and reduce variance between folds. The detailed performance metrics shown Table 4.

VGG16, despite being an older architecture, proved to be the most stable model in this study. Although its accuracy (95.49%) is slightly lower than InceptionV3, its minimal variance (± 0.08) implies that VGG16 is exceptionally resistant to perturbations in the training data. The detailed performance metrics shown Table 5.

ViT-B-16 achieved a competitive accuracy of 94.41%. While successful, it slightly lagged behind the hierarchical Swin Transformer, suggesting that the hierarchical approach of SwinT is better suited for extracting local features in this domain compared to the standard Vision Transformer. The detailed performance metrics shown Table 6.

Table 2 Comparing the Performance of the Models

Model	Accuracy	Precision	Recall	F1 Score
InceptionV3	0.9643	0.9645	0.9643	0.9642
ResNet50	0.9634	0.9637	0.9634	0.9633
VGG16	0.9558	0.9562	0.9558	0.9558
SwinT	0.9608	0.9609	0.9608	0.9607
CheXNet	0.9586	0.9588	0.9586	0.9586
ViT-B16	0.9439	0.9445	0.9439	0.9438

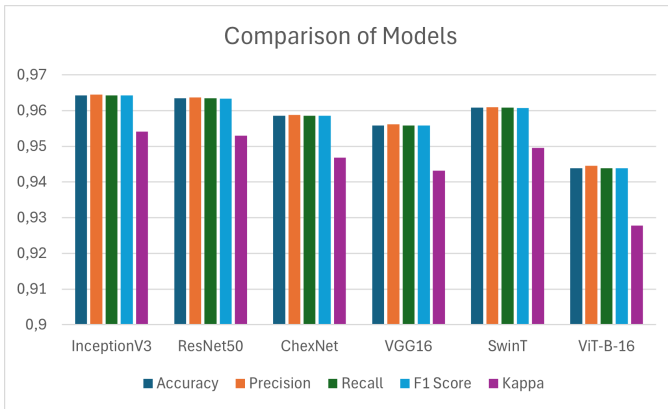


Figure 2 Model performance comparison on the multi-class classification dataset.

B. Application Outputs

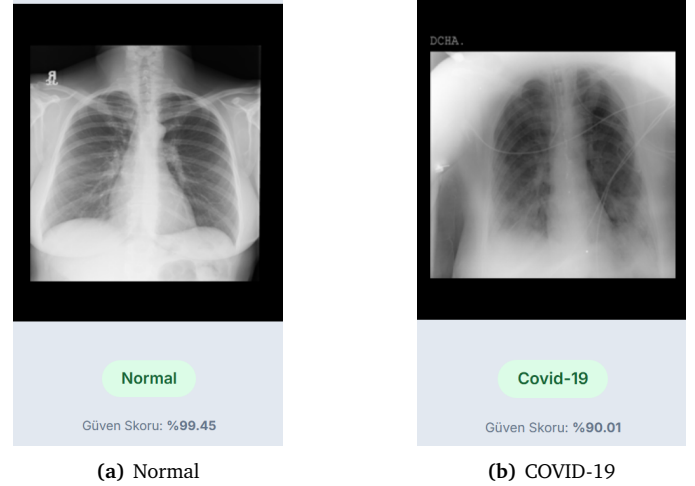


Figure 3 Example of InceptionV3 Application Results

C. Conclusion

This study presented a comparative analysis of state-of-the-art CNN (InceptionV3, ResNet50, VGG16, CheXNet) and Transformer (SwinT, ViT) architectures for the multi-class classification of lung diseases.

Based on the experimental results, InceptionV3 is identified as the optimal model for this specific task, offering the highest accuracy %96.43 and F1 score. However, the Swin Transformer proved to be a powerful alternative, achieving results statistically equivalent to InceptionV3 %96.42, thereby validating the efficacy of attention-based mechanisms in medical diagnostics.

While ResNet50 and VGG16 offered slightly lower accuracy, they demonstrated superior stability, particularly VGG16, which had the lowest variance among all models. Conversely, CheXNet exhibited significant instability and lower generalization capability on this dataset, highlighting the importance of cross-validation in detecting model reliability issues.

In conclusion, while CNNs currently maintain a slight edge in accuracy, hierarchical Transformers like SwinT are rapidly closing the gap. Future work may focus on hybrid architectures that combine the feature extraction stability of CNNs with the global context awareness of Transformers to further enhance diagnostic precision.

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